

Deepak Mallampati

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Kent, OH (US-based, F-1 OPT, open to sponsorship)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) who translates business needs into Python-backed LLM applications, multi-provider REST API integrations, and stakeholder-facing system-limitation docs. Production-shipped: Healthcare RAG (~35% retrieval precision), agentic LLM workflows (>30% hallucination reduction), Manga Lens (Chrome Web Store, four AI vision providers, 29 site configs). FastAPI / Docker / Jenkins CI/CD production discipline with ~30% post-deployment defect reduction. Open to an engineering-to-Solutions-Engineer transition with structured CCaaS product ramp.

SKILLS

REST API Integration • Multi-Provider AI Communications • LLM Application Delivery • Stakeholder Communication • System-Limitation Documentation • Python / FastAPI / Docker • CI/CD Pipelines • ROI & Cost Analysis

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

LLM & GenAI: LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Integration & APIs: RESTful APIs, multi-provider API orchestration (Claude / OpenAI / Ollama in Manga Lens; Perplexity Sonar + Replicate in Dream Decoder), JSON contracts, structured logging

Infra & DevOps: Docker, Jenkins CI/CD, Grafana, observability, MLOps/LLMOps, cloud-ready deployment

Domains: Healthcare AI, Applied AI, Enterprise AI, Workflow Automation, Multimodal Systems

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA (Healthcare Technology, AI Consulting)

- Wrote **stakeholder-facing system-limitation docs** alongside every model release - the artifact that earned trust faster than improving the model itself; trained clinicians and ops on inputs/outputs/refusal cases.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, structured logging + load simulation; post-deployment defects **~30%**.
- Developed **agentic LLM workflows** with structured reasoning, tool discipline, grounding rules; agent stability **+~25%**, hallucination reduction **>30%**.
- Built **RAG systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings; contextual retrieval precision **+~35%**.

Kent State University

Aug 2023 - May 2025

Student Manager - Ohio, USA (Higher Education, Operations Management)

- Led coordination for **150+ students**; structured scheduling and communication reduced coordination delays **~30%** and logistical issues **~25%**.
- Managed event planning, **budget tracking, and vendor comparison**; achieved **10-15% cost efficiency** in select initiatives without quality loss - ROI/TCO-shaped trade-off conversations with non-technical stakeholders.
- Built attendance, participation, and feedback tracking in Excel; surfaced engagement trends that informed scheduling decisions.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India

- Built **CI/CD pipelines with Jenkins**; deployment errors **->30%**, release cycle **-35-40%**.
- Designed performance dashboards (**SQL Server DMVs and Grafana**); incident recurrence **~25%**.
- Supported RBAC implementation and audit logging for financial modules in a compliance-sensitive environment.

PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation (shipped, Chrome Web Store)

Full-stack browser extension built independently - Manifest V3, content scripts, service workers, and `captureVisibleTab`. Integrates **four AI vision providers** via REST (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local) with per-provider payload handling. Multi-section capture pipeline; 7-day cache; 29 site configs; privacy policy; narrowed host permissions. **Direct multi-provider REST integration evidence.**

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs (Perplexity Sonar for interpretation + Replicate image models for 16:9 visuals). **Intermediate structured prompt transformation layers** improved contextual alignment ~**30%** and first-pass image success rate ~**25-30%**.

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline with **schema-validated JSON contracts between agents**. RAG-grounded CPT/ICD validation; ANN duplicate detection; explainable risk scoring with audit-ready reasoning traces - delivery archetype for regulated-domain solution work.

Driver Drowsiness Detection - Real-Time YOLOv8 Fatigue Monitoring

Replaced a two-stage CNN pipeline with a **unified YOLOv8** model; inference latency ~**30%**, false positives ~**25%** via sliding-window confidence aggregation.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025